

# Homework — 1.3.3 Networks

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Name: \_\_\_\_\_ Date: \_ **Mark:** \_\_\_\_ / 20

OCR H446 — set after the 1.3.3 lesson. Show your working.

## Part A — This week's topic (~14 marks)

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1. State **two** differences between a **LAN** and a **WAN**. [2] (AO1)

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2. A small office is connected using a **star topology** built around a central switch.

(a) State **one** advantage of a star topology. [1] (AO1)

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(b) State **one** disadvantage of a star topology. [1] (AO1)

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3. Explain how the **Domain Name System (DNS)** allows a user to reach a web server after typing a domain name into their browser. [3] (AO1/AO2)

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4. A company sets up a **client-server** network instead of a peer-to-peer network. Give **two** advantages and **one** disadvantage of this choice. [3] (AO2)

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5. A user retrieves their email. Explain the difference between **POP3** and **IMAP** for retrieving mail, and name the protocol used to **send** mail. **[3]** (AO1)

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6. Describe what a **firewall** does to help protect a network. **[1]** (AO1)

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## Part B — Retrieval: keep it fresh (~6 marks)

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7. (1.3.2) State what is meant by **referential integrity** in a relational database. **[2]** (AO1)

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8. (1.3.1) State **two** properties that a good hash function should have. **[2]** (AO1)

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9. (1.2.2) Explain **one** task carried out by a **linker** when an application is built. **[2]** (AO1)

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